

e/m by J.J. Thomson Method

OBJECT:- To determine the value of e/m of electron by J.J. Thomson Method.

APPARATUS: CRT tube, deflection set up, bar magnet, tangent galvanometer and wooden stand.

FORMULA:-

$$\frac{e}{m} = \frac{yV}{LldB^2}$$

Where y = deflection in CRT tube

V = deflecting voltage in volts

L = distance from centre of deflection plates to screen

l = length of deflection plates

d = distance between plates

B = applied magnetic field

SET UP SPECIFICATIONS:-

L = 17 cm, l = 2 cm, d = 0.4 cm & B_H (Horizontal component of Earth magnetic field) = 3×10^{-5} Wb./Sq.m (Tesla)

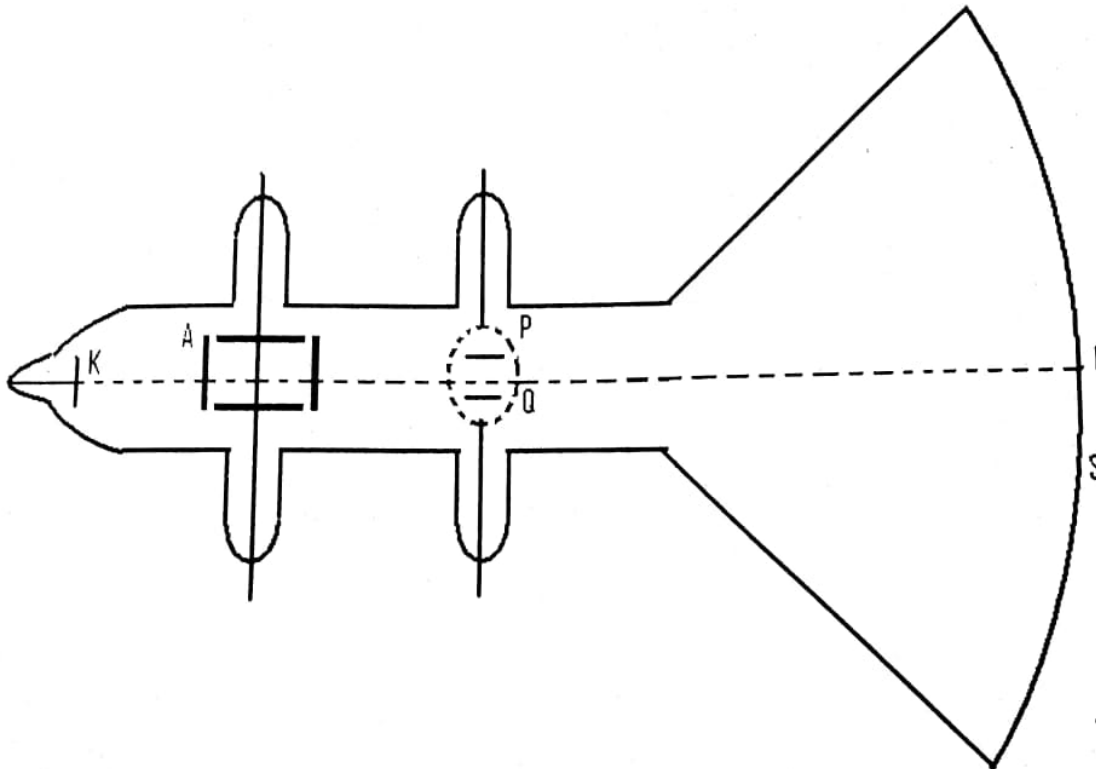
THEORY:-

The apparatus consists of a CRT which has three essential parts.

- i). Electron gun: This part consists of the cathode K and the anode A. It produces accelerates and focuses the electrons into a fine beam so as to produce a small bright spot on the fluorescent screen S.
- ii). Screen: The screen S is coated with some fluorescent material like Zinc Sulphide etc., so that a greenish blue spot of light is observed where the electron beam impinges

upon the screen. A transparent mm graph is attached to the screen to note the exact position of the fluorescent spot.

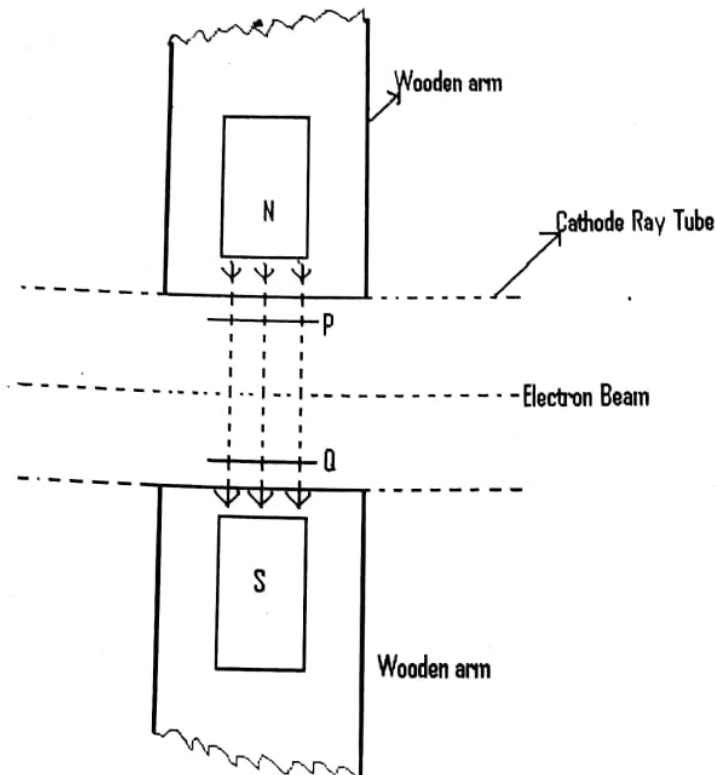
iii). Electric and Magnetic deflecting system: Two plates P and Q are fitted in the tube symmetrically on other side of the electron beam so that an electric field perpendicular to the plane the paper can be applied between them.



A magnetic field can also be applied at the same place in a direction perpendicular to the direction of the electric field i.e. in the plane of the paper. Thus the deflection of the cathode ray due to the electric as well as the magnetic field takes place in a direction perpendicular to the plane of the paper.

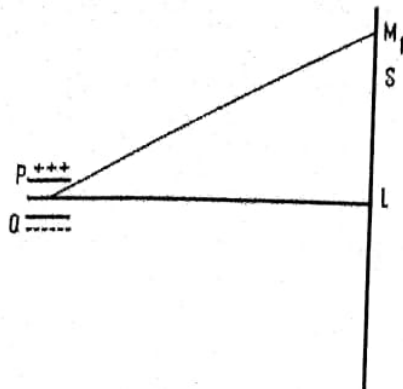
To apply the magnetic field, the cathode ray tube is fitted on a wooden stand provided with two arms, one on either side perpendicular to the direction of the electron beam. A scale is marked on each arm which gives the distance from the axis of the cathode ray tube. When two bar magnets are placed with their lengths parallel to the arms with their opposite poles towards the cathode ray tube, these produce a uniform magnetic

field exactly at the same place where an electric field is produced when a potential difference is applied between the plates P and Q.

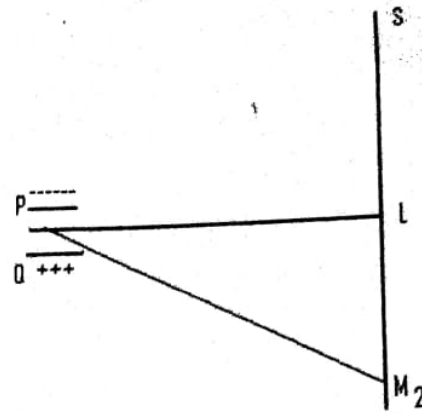


Let E be intensity of the electric field applied between the plates P and Q in a direction perpendicular to the plane of the paper then the force acting on the electron of charge $= Ee$

This force acts at right angle to the direction of motion of the electron. The electron, therefore, moves along a circular path within the electric field and on leaving the field flies off tangentially to the path meeting the screen at M_1 or M_2 depending upon the direction of the electric field as shown in figure below. When a magnetic field B is applied in the plane of the paper perpendicular to the direction of the beam as shown figure a then according to Fleming's left hand rule the beam is deflected upward out of the paper in a direction perpendicular to both, the direction of motion of the electron and that of the field.

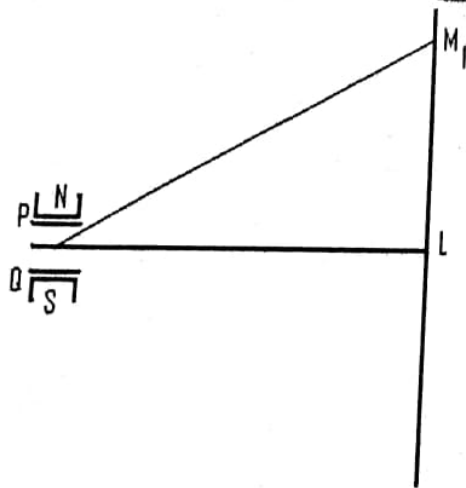


(a)

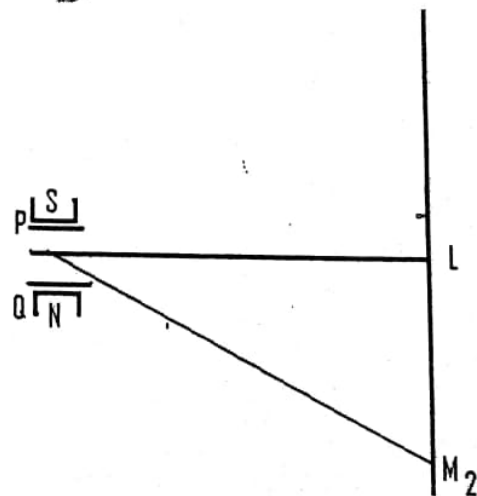


(b)

When a magnetic field is applied as shown in figure above (b) the electron beam is deflected, downward into the paper. If v is the velocity of the electrons in the beam, then magnitude of the force acting on each electron $= Bev$. As this force also acts at right angles to the direction of motion of the electron, the electron moves along a circular path in the magnetic field and on leaving the field flies off tangentially meeting the screen at M_1 or M_2 as explained above.



(a)



(b)

If r is the radius of the circular path along which the electron moves in the magnetic field.

$$Bev = \frac{mv^2}{r}$$

OR

$$\frac{e}{m} = \frac{v}{rB} \text{ -----(i)}$$

To find the velocity of electron v: To find the value of v , one of the fields say the magnetic field is first applied. The spot of light moves from its initial position to the position say M_1 . The electric field is now applied simultaneously so as to produce deflection of the electron beam in the opposite direction. The value of the electric field is so adjusted that the spot of light comes back to its initial position. In such a case the forces acting on the electron due to the electric and magnetic fields are equal and opposite.

$$Ee = Bev \quad \text{OR} \quad \frac{E}{B} = v$$

Substituting in (i), we have

$$\frac{e}{m} = \frac{E}{rB^2} \dots \dots \dots (ii)$$

To find the radius r of circular path: To find the value of r , the radius of the circular path in the magnetic field, note the position of the luminous spot on the screen when no field is applied and when a field B is applied. Let the deflection of the spot be $y = LM_2$ as shown. The circular path of the beam OR in the magnetic field is as shown in above figure. The beam leaves the magnetic field at R tangentially to the path and meets the screen at M_2 so that M_2R produced meets OX in T . Draw OC and RC perpendiculars to OL and RM_2 respectively meeting at C , is the centre of the circular arc OR and $OC = CR =$ radius of the circle $= r$

Now

$$\angle OCR = \angle LTM_2 = \theta$$

$$\tan \theta = \frac{LM_2}{LT} = \frac{OX}{OC}$$

$$r = OC = \frac{LT(OX)}{LM_2}$$

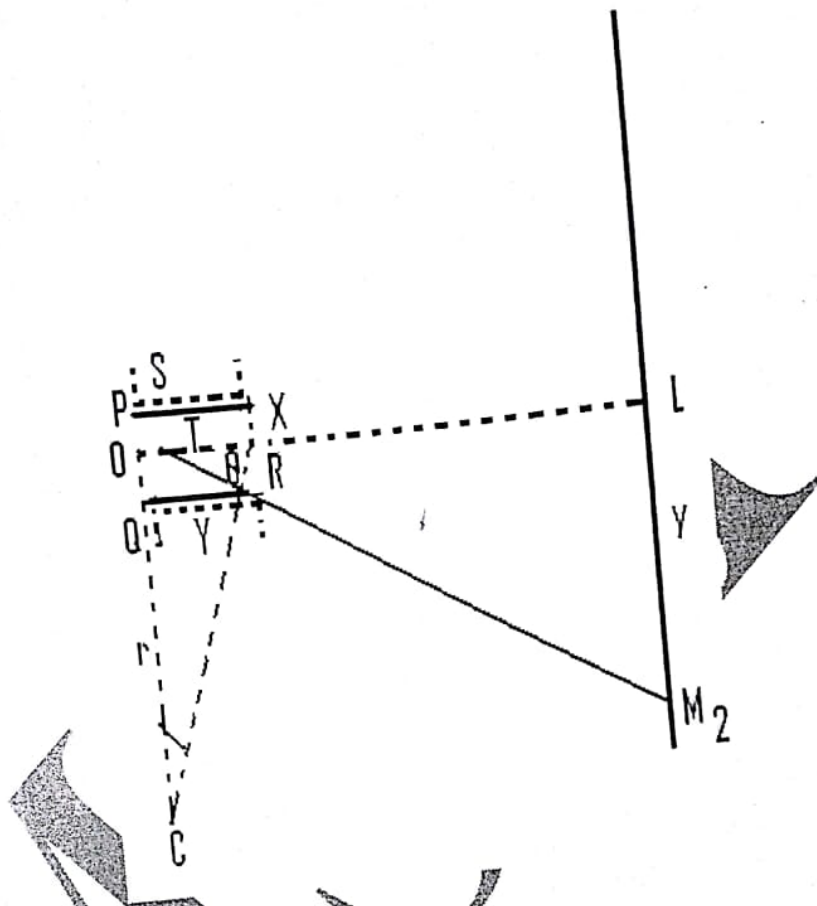
Now OX is very nearly equal to the length of the region in which the magnetic field is applied. This is also equal to the length of the region in which the electric field is

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applied i.e. the length of the plates P and Q. Let it be $= l$ (This is generally given by the makers).



(T is the distance of the centre of the screen from the centre of the region of the magnetic field & electric field). Let it be $= L$

$$r = \frac{Ll}{y}$$

Substituting in (ii), we have

$$\frac{e}{m} = \frac{yE}{LlB^2} \dots\dots\dots(iii)$$

The electric field $E = V/d$ where V is the potential difference applied between the plates P and Q and d is the distance between the plates (d is also given by the makers). Substituting in (iii), we have

$$\frac{e}{m} = \frac{yV}{LldB^2}$$

When V is in volts, B in Tesla and y , L , l and d in meters e/m is in coulomb/kg

PROCEDURE:-

1. Draw the north-south line using a compass needle. Place the cathode ray tube fitted in the wooden frame with its axis along the north-south line so that the arms of the frame lie along the east-west line.
2. Connect the cathode ray tube to the power supply unit. Switch on the current and wait till a luminous(greenish) bright spot appears on the screen. Adjust the (intensity) brightness and focus controls so as to get a sharp bright spot point in the middle of the screen. Note the initial position of the spot on the scale fitted on the screen.
3. Now apply a suitable deflecting voltage so that the luminous spot is deflected by about 0.5 cm to 1.0 cm. Note the deflecting voltage V and the position of the spot. Measure the distance through which the spot has removed and let it be y (use the diff. cont. knob to deflect the spot)

4. Place the bar magnets symmetrically on either side of the cathode ray tube along the arms of the wooden stand on which the tube is fitted such that their opposite poles face each other and their common axis is exactly at right angles to the axis of the cathode ray tube. Adjust the polarity as well as the distance of the luminous spot comes back to its initial position. When the adjustment is perfect note the distance of the poles of the magnets on the side nearer to the cathode ray tube. Let the distance be r_1 (The magnets should be placed 17 cm from the screen)
5. Remove the bar magnet, switch off the electric field applied to the deflecting plates and again note the initial position of the luminous spot. Reverse the polarity of the difference applied to the electric deflecting plates with the help of the reversing switch fitted in the power supply unit thereby reversing the electric field. Again note the final position of the luminous spot and calculate y . Again place the bar magnets on the arms of the wooden stand as in the previous step and adjust their polarity as well as the distance so that spot comes back to its initial position. When the adjustment is perfect again note the distances of the poles of the magnets on the side nearer to the CRT. Let the distance be r_1 . Switch off the power supply.
6. To find the value of the magnetic field B , carefully remove the magnets and the CRT from the wooden stand. Place the compass box (of a deflection magnetometer or tangent galvanometer) such that its centre lies exactly on the point where the common axis of the bar magnets

Note: The axis of the CRT intersect. Rotate the compass box about the vertical axis so that the pointer lies along the 0 - 0 line. Place the magnets exactly in the same positions as in step 4 at distance r_1 . This produces a deflection in the magnetometer compass box and the two ends of the pointer give the deflection. Let the reading be θ_1 and θ_2 .

$$\theta = \frac{\theta_1 + \theta_2}{2}$$

Now place the magnets exactly in the same positions as in step 5 at distances r_1 and again note the deflection θ_1 and θ_2 from the two ends of the pointer of the compass box. Then

$$\theta = \frac{\theta_1 + \theta_2}{2}$$

If B_H is the horizontal component of earth's magnetic field. Then

$$B = B_H \tan \theta$$

Take one more sets of observations by changing the value of 'V' and hence that of the electric field.

OBSERVATION TABLE:-

S. No	Applied Voltage	Direct Field				Reverse Field					
		Position of spot		Deflection	Magnetic pole distance		Position of spot			Magnetic pole distance	
		Initial	Final				Initial	Final	Deflection		
				y_D	North	South			y_R	North	South
1.	1.2	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5		
2.											
3.											

DETERMINATION OF B:-

S. No	Applied Voltage V	Reading of two ends of the pointer when magnetic poles at r ₁ (Direct Field)			Direct field B=B _H tanθ		Reading of two ends of the pointer when magnetic poles at r ₁ (Reverse Field)			Direct field B=B _H tanθ	
		θ ₁	θ ₂	Avg. θ = θ ₁ + θ ₂ /2			θ ₁	θ ₂	Avg. θ = θ ₁ + θ ₂ /2		
1.											
2.											
3.											

CALCULATIONS:-

$$\text{i). } \frac{e}{m} = \frac{yV_1}{Ll dB^2} = \text{C/kg (Direct Position)}$$

$$\text{ii). } \frac{e}{m} = \frac{yV_1}{Ll dB^2} = \text{C/kg (Reverse Position)}$$

$$\text{iii). } \frac{e}{m} = \frac{yV_2}{Ll dB^2} = \text{C/kg (Direct Position)}$$

$$\text{iv). } \frac{e}{m} = \frac{yV_2}{Ll dB^2} = \text{C/kg (Reverse Position)}$$

Average $e/m =$

RESULT: -

PERCENTAGE ERROR:-

PRECAUTIONS:-

1. The CRT must be set in the north-south direction.
2. The luminous spot on the screen should be sharp, bright and in the middle.
3. The deflecting voltage must produce a deflection of the order of 0.5 – 1.0 cm.
4. The CRT must be handled carefully.
5. The deflecting voltage must be reversed to produce the deflection of the spot of light in the opposite directions and the magnetic field must also be reversed to bring it back to initial position.
6. The axis of the bar magnets and the axis of the CRT must be exactly at right angles and in a horizontal plane.

Viva-Voce

- Q.No.01 Can you mention some other methods of finding e/m ?
- Q.No.02 Why is the CRT axis placed along the magnetic meridian?
- Q.No.03 What is the nature of the electron path in the electrostatic field?
- Q.No.04 What is the nature of electron path in the magnetic field?
- Q.No.05 Can the earth's magnetic field be used to deflect the beam in your experiment?
- Q.No.06 How can the mass of an electron be determined?
- Q.No.07 What are cathode rays?